

All About State Machines: The Smart Traffic Light

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1 What are Finite State Machines?

Imagine a toy that has different "moods." A robot dog, for example, can "Sleep," "Bark," or "Run." But it can never do everything at the same time. This is called a **Finite State Machine (FSM)**.

- **Finite** means: There is a fixed number of moods (states). It cannot suddenly do something completely unknown.
- **State** means: What the machine is doing right now.

2 Describing Machines: State Chart Diagrams

To understand how the robot thinks, we draw a **State Chart Diagram**. It is like a map with cities and roads.

- **Circles (Nodes):** These are the **states** (e.g., "Light is Red"). They are the basic elements representing a specific condition.
- **Arrows (Edges):** These are the **transitions** or paths from one state to the next.
- **Events (Inputs):** This is the text written on the arrows. it is the reason why the machine moves (e.g., "Button pressed").

3 Moore vs. Mealy: Two Types of Robots

There are two main ways machines handle their tasks:

1. **Moore Machine:** The machine only looks at its current mood. If it is in the "Happy" state, it laughs. It doesn't matter what is happening outside at that exact moment. The **output depends only on the state**.
2. **Mealy Machine:** This machine reacts faster. It laughs only if it is in the "Happy" state **and** someone tells a joke at the same time. The **output depends on both the state and the input**.

4 The Traffic Light Example

Here, our machine controls a traffic light for cars that reacts to a pedestrian button (X).

4.1 State Chart Diagram

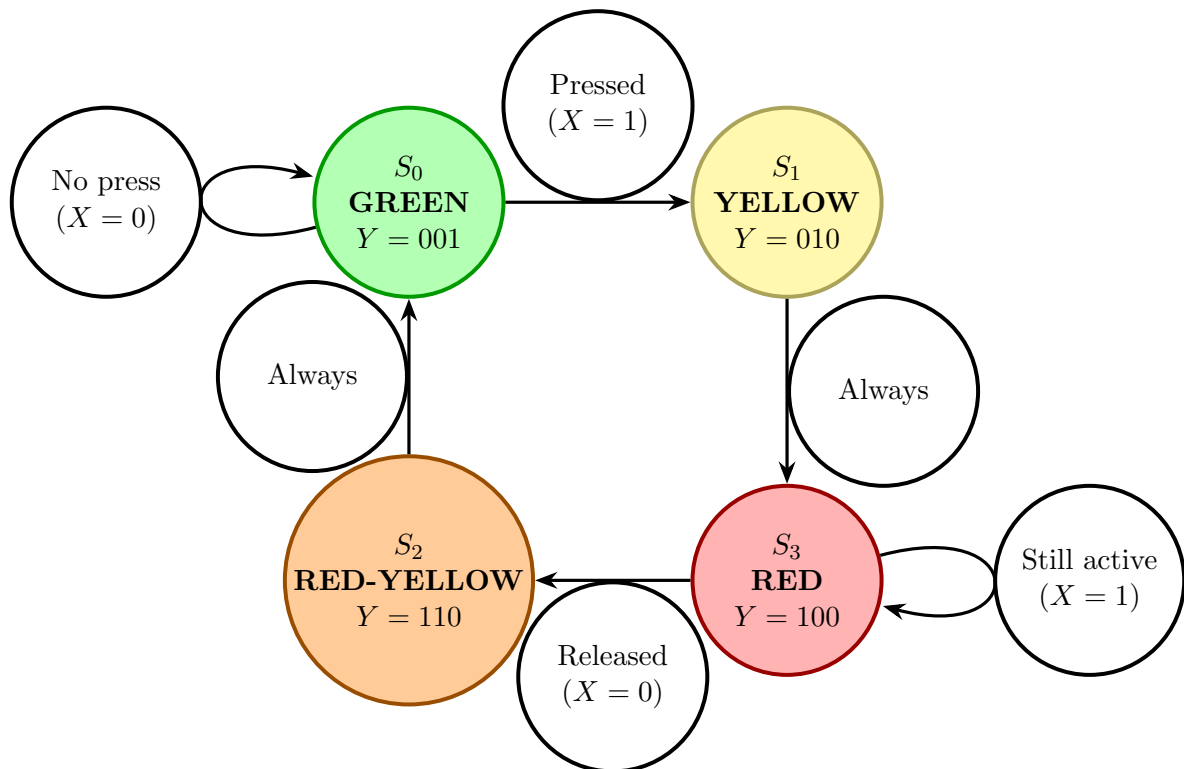


Figure 1: Moore state machine for traffic light control[cite: 30, 36].

5 Implementing the Moore Machine

To build the machine, we fill in two functional tables.

5.1 1. Transient Functions (Transitions)

This table tells us: "If I am here and this happens, I go there."

Current State	Input (Button X)	Next State	Explanation
S_0 (Green)	0	S_0	No one presses, stays Green.
S_0 (Green)	1	S_1	Someone presses, turns Yellow.
S_1 (Yellow)	0 or 1	S_3	Yellow always turns to Red.
S_3 (Red)	1	S_3	Stays Red as long as pressed.
S_3 (Red)	0	S_2	Button released, preparation.
S_2 (Red-Yellow)	0 or 1	S_0	Back to start (Green).

5.2 2. Output Functions

This table tells us: "In this state, these lights turn on." (y_2 =Red, y_1 =Yellow, y_0 =Green).

State	Code ($y_2y_1y_0$)	Light Signal	What does the car see?
S_0	001	Green	Driving allowed.
S_1	010	Yellow	Caution, stop soon.
S_2	110	Red + Yellow	Caution, starting soon.
S_3	100	Red	Stop!